

TfL “What-If” Simulator User Story Document

Must-Have User Stories:

US-01 – View the Underground Network Map

As a user, I want to view an interactive map of the London Underground, **so that I** can understand how stations and lines are connected.

- **Acceptance Criteria:**
 - The map displays all underground lines – connecting stations.
 - Each line is shown using its official TfL colour.
 - Users can pan and zoom the map.
 - Stations are clearly visible and selectable.
- **Priority** – Must Have

US-02 – Accurate Station Hierarchy Representation

As a transport analyst, I want stations to be grouped into hubs with their platforms and entrances, **so that** the simulation reflects the real Underground structure.

- **Acceptance Criteria:**
 - Parent stations represent real-world hubs.
 - Child nodes represent platforms or entrances.
 - Connections between nodes match TfL data.
 - No orphan or disconnected stations appear.
- **Priority** – Must Have

US-03 – Calculate the Shortest Route Between Stations

As a commuter, I want to calculate the shortest route between two stations, **so that I** can see the fastest route to travel across the network.

- **Acceptance Criteria:**
 - Users can select a start and destination station.
 - System calculates and displays the shortest route.
 - Travel time is shown for the route.
 - Routes must update when network conditions change.
- **Priority** – Must Have

US-04 – Test “What-If” Disruption Scenarios

As a planner, I want to manually close stations or lines, **so that I** can analyse the impact of disruptions.

- **Acceptance Criteria:**
 - Users can toggle stations as “Open” or “Closed”.
 - Closed stations are removed from routing.
 - Users can also disrupt certain lines causing rerouting.
 - Alternative routes are recalculated instantly.
 - Changes are visually reflected on the map.
- **Priority** – Must Have

US-05 – React to Live Service Disruptions

As a user, I want the simulation to reflect real-time disruptions, **so that** routes and travel times remain realistic.

- **Acceptance Criteria:**
 - Closed or disrupted lines are visually indicated.
 - Affected routes are recalculated automatically.
 - Closed stations cannot be routed through.
 - Updates occur without requiring a page refresh.
- **Priority** – Must Have

Nice-To-Have User Stories

US-06 – Simulate Crowding and Overcapacity

As a transport planner, I want to see how overcrowding affects stations, **so that I** can assess capacity issues during peak times.

- **Acceptance Criteria:**
 - Stations display congestion visually.
 - Overcrowded stations cause delays.
 - Queues or reduced flows are shown.
 - Crowding levels update dynamically.
- **Priority** – Nice to Have

US-07 – Measure Network Resilience

As an analyst, I want to measure how resilient the network is to disruptions, **so that I** can identify critical stations and bottlenecks.

- **Acceptance Criteria:**
 - Average travel time is calculated.
 - Travel time changes when hubs are closed.
 - Results are shown numerically or visually.
 - Major hubs can be compared against each other.
- **Priority** – Nice to Have

US-08 – Enhanced 3D Visualisation

As a user, I want a 3D visual representation of the network, **so that** the simulation is more engaging and intuitive.

- **Acceptance Criteria:**
 - Lines or stations have 3D depth.
 - Visuals remain clear and uncluttered.
 - Performance remains smooth.
 - Users can still pan and zoom easily.
- **Priority** – Nice to Have